ModelIt!

Student Activity Pack

Why Is It Dark at Night?

The Science of Day and Night Patterns

Grade 1 | NGSS 1-ESS1-1 | Patterns

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 Draw what the sky looks like during the day and what it looks like at night.

2 Circle the correct answer: The sun goes away at night / The Earth spins and our side faces away from the sun.

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Earth's Position

Amount of Sunlight

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Facing the Sun

Earth's Position set to facing the sun

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Facing Away

Earth's Position set to facing away from the sun

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Why does the sky get dark at night and bright during the day? Use evidence from your simulation to support your answers.

8 STEM Challenge: Build a Day and Night Model

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YOUR ENGINEERING MISSION

Create a model showing how the Earth spinning causes day and night using a ball and a flashlight.

THE SCENARIO

A young student from another class asked you: Why does the sun go away at night? You know the answer — the sun does NOT go away! Build a model to show this younger student how day and night really work.

Engineering Design Thinking

1. WHAT ROUND OBJECT WILL YOU USE FOR THE EARTH?
2. HOW WILL YOU SHOW WHICH PART OF THE EARTH HAS DAYTIME?
3. HOW WILL YOU SPIN THE EARTH TO SHOW DAY CHANGING TO NIGHT?
4. CAN YOU MARK A SPOT ON YOUR EARTH TO SHOW WHERE YOU LIVE AND TRACK WHEN IT IS DAY AND NIGHT?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Astronomers study the sun, moon, stars, and planets to understand how they move. They use telescopes and computers. They earn $60,000-$120,000/year.